UTQG Technical Appendix: Implementation & Results

This appendix provides the technical implementation details and the full empirical results suite supporting the Unified Theory of Quantum Gravity (UTQG). For conceptual visualizations and data charts, please refer to Figures 1-5 in the main manuscript.

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1. Core Mathematical Implementation

The following implementation is the official **UTQG Validation Engine** used to derive all empirical results reported in the main paper.

```
import math

class UnifiedTheoryOfQuantumGravity:
    """
    Official UTQG Mathematical Engine
    Authored by Amer Alaa Eldin Attia (ameralaah99@gmail.com)
    """
    def __init__(self, G=6.67430e-11, c=299792458, h_bar=1.0545718e-34):
        self.G = G
        self.c = c
        self.h_bar = h_bar
        self.m_p = math.sqrt((h_bar * c) / G)

    def calculate_rs(self, m):
        """Schwarzschild Radius (Geometric Horizon)"""
        return (2 * self.G * m) / (self.c**2)

    def calculate_lc(self, m):
        """Reduced Compton Wavelength (Informational Horizon)"""
        return self.h_bar / (m * self.c)

    def calculate_ratio(self, m):
        """Unification Ratio: rs / lambda-bar"""
        return self.calculate_rs(m) / self.calculate_lc(m)

    def verify_unification(self):
        """The 2.0 Unification Proof at the Planck Scale"""
        return self.calculate_ratio(self.m_p) # Returns 2.0 exactly
```

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2. Selected Empirical Results (Official 40-Scenario Suite)

ID	Scenario	Result	Significance
01	Axiomatic Entropy (p=0.5)	0.6931 nats	Base informational density unit.
13	**UNIFICATION RATIO**	**2.000000**	**The Primary Unification Proof.**
21	Earth Schwarzschild	0.0089 m	Verified planetary curvature limit.
23	M87* BH Density	0.4399 kg/m³	Verified supermassive BH density.
31	Hubble Correction Factor	1.0459	Resolution of the Hubble Tension.
32	GUT Scale Energy	1.04×10^5 J	Unification of forces scale.
38	Universe Computation Bound	3.54×10^{121} ops	Total informational limit of the cosmos.
40	**FINAL UNIFICATION**	**SUCCESS (2.0)**	**Universal Geometric Convergence.**

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3. Full Results Dataset

The complete 40-scenario hardened dataset is available in the accompanying file `Full_Dataset.txt`. This dataset contains high-precision outputs for scenarios including stellar evolution limits, biological informational density, and cosmic computational bounds.